

Inputs from encoder

Family representation
 r^h

Member representations
 $\{r_n\}_{n=1}^N$

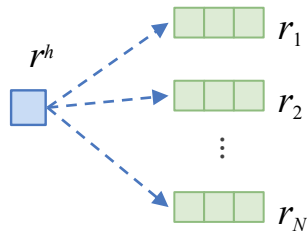
Household-level pattern prior
 π^h

Individual-level pattern priors
 $\{\pi_n^{ind}\}_{n=1}^N$

Stage 1 Coordinated member context

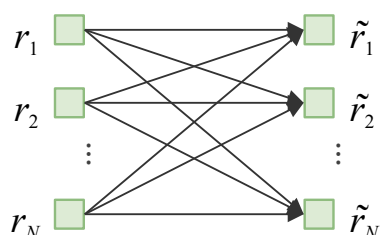
1. Task-specific attention

Each member attends to family representation



2. Cross-role attention

Members interact with co-members



Coordinated member contexts

$\{\tilde{r}_n\}_{n=1}^N$

$\tilde{r}_1$ [blue rectangle]

$\tilde{r}_2$ [blue rectangle]

$\vdots$

$\tilde{r}_N$ [blue rectangle]

Stage 2 Autoregressive activity-chain generation

Decoding over steps $t = 1, 2, \dots, T$

Decoder input for member n at step t

Embed($a_{n,t-1}$)
(previous activity embedding)

+ PosEmbed(t)
(position embedding)

+ $\tilde{r}_n$
(coordinated member context)

Stacked Transformer Decoder Layer ($\times L$ layers)

1. Causal self-attention
(over past activities)

2. Cross-attention to r_n
(member representation)

3. Adaptive layer normalization
(conditioned on π^h, π_n^{ind})

4. Feed-forward network (FFN)

Dedicated prediction heads (for member n at step t)

Departure time

Arrival time

Purpose

Travel mode

Driver flag

Joint-travel flag

Destination

Spatially-aware destination attention

Nash-bargaining coordination layer

Enforce household feasibility and maximize household utility

Phase 1

Obtain a feasible converged solution for this game using sequential quadratic programming

Phase 2

Obtain the backward gradients of the converged solution with respect to model parameter